

Zener diode

Zener diode

Zener diode is a special type of device designed to operate in the zener breakdown region (*reverse breakdown region*). The breakdown voltage of a zener diode is carefully set by controlling the doping level during manufacture.

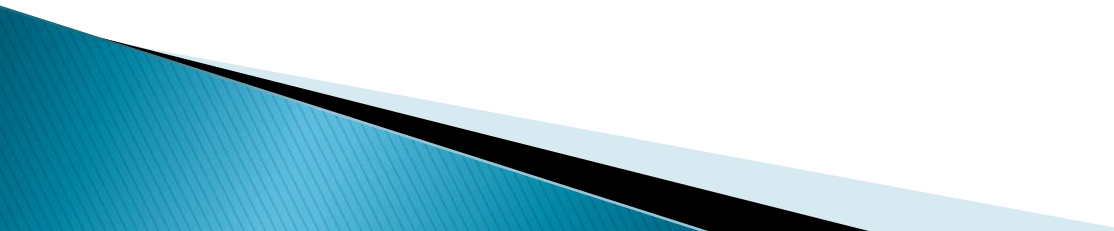
If the diode is heavily doped, zener breakdown occurs at low reverse voltages. On the other hand, if the diode is lightly doped, the zener breakdown occurs at high reverse voltages. Zener diodes are available with zener voltages in the range of 1.8V to 400V.

Zener diode is heavily doped than the normal p-n junction diode. Hence, it has very thin depletion region. Therefore, it allows more electric current than the normal p-n junction diodes.



Zener diode allows electric current in forward direction like a normal diode but also allows electric current in the reverse direction if the applied reverse voltage is greater than the zener voltage. So, it is always connected in reverse direction because it is specifically designed to work under reverse bias.

Zener diodes are the basic building blocks of electronic circuits and widely used in all kinds of electronic equipments to protect electronic circuits from over voltage.



Symbol of zener diode

The symbol of zener diode is shown in figure. It consists of two terminals: cathode and anode. In zener diode, electric current flows from both anode to cathode and cathode to anode. The symbol of zener diode is similar to the normal p-n junction diode, but with bend edges on the vertical bar.



Breakdown in zener diode

There are two types of reverse breakdown regions in a zener diode:

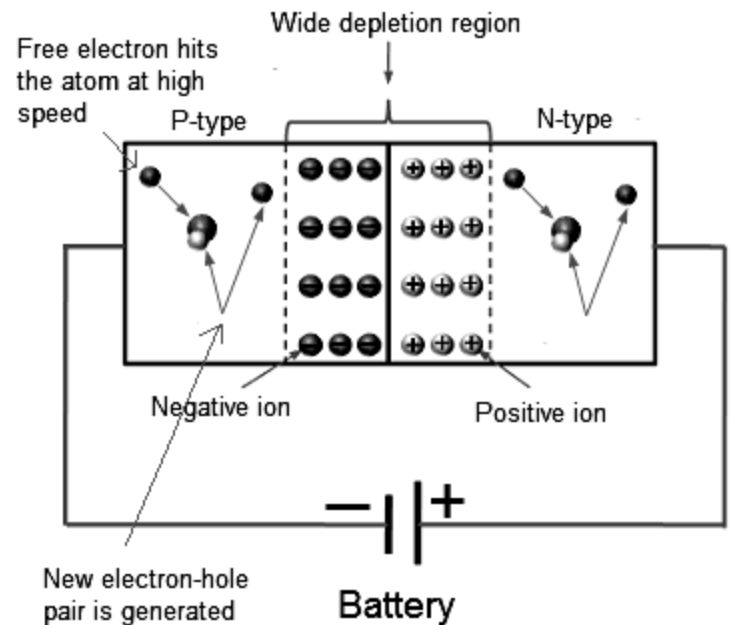
avalanche breakdown

zener breakdown.

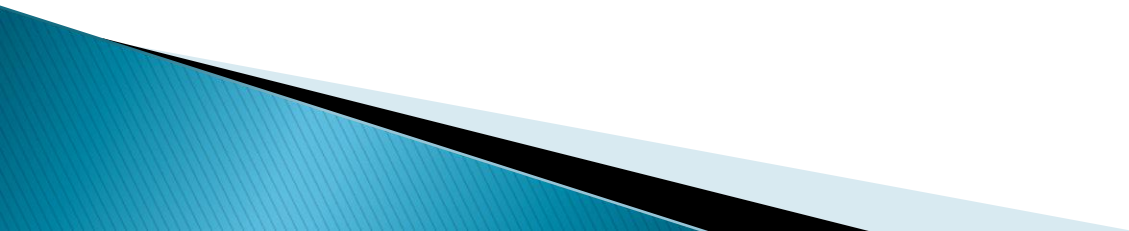
Avalanche breakdown

The avalanche breakdown occurs in both normal diodes and zener diodes at **high reverse voltage**. When high reverse voltage is applied to the p-n junction diode, the free electrons gain a large amount of energy and are accelerated to greater velocities.

The free electrons moving at high speed will collide with the atoms and knock off more electrons. These electrons are again accelerated and collide with other atoms. Because of this continuous collision with the atoms, a large number of free electrons are generated. As a result, electric current in the diode increases rapidly. This sudden increase in electric current may permanently destroy the normal diode.

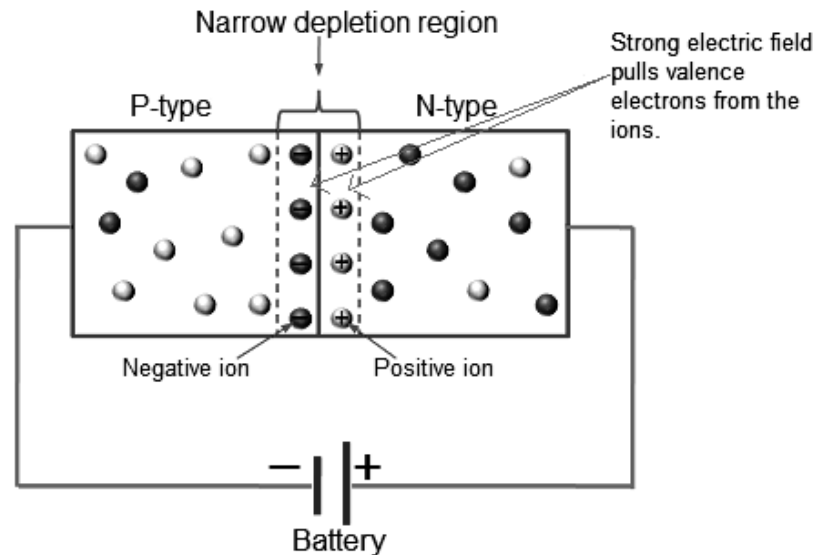


However, avalanche diodes may not be destroyed because they are carefully designed to operate in avalanche breakdown region. Avalanche breakdown occurs in zener diodes with zener voltage (V_z) greater than 6V.

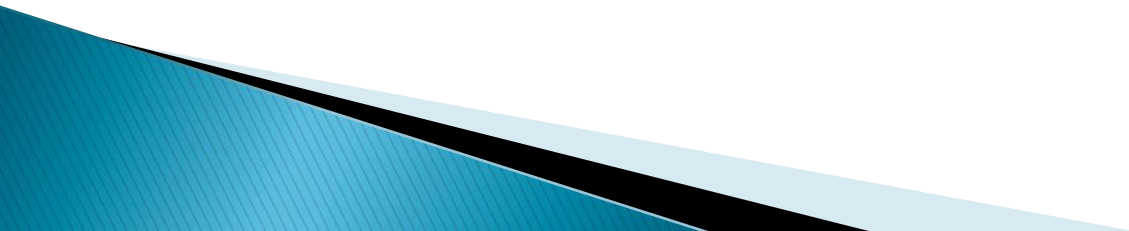


Zener breakdown

The zener breakdown occurs in heavily doped p-n junction diodes because of their narrow depletion region. When reverse biased voltage applied to the diode is increased, the narrow depletion region generates strong electric field. When reverse biased voltage applied to the diode reaches close to zener voltage, the electric field in the depletion region is strong enough to pull electrons from their valence band.

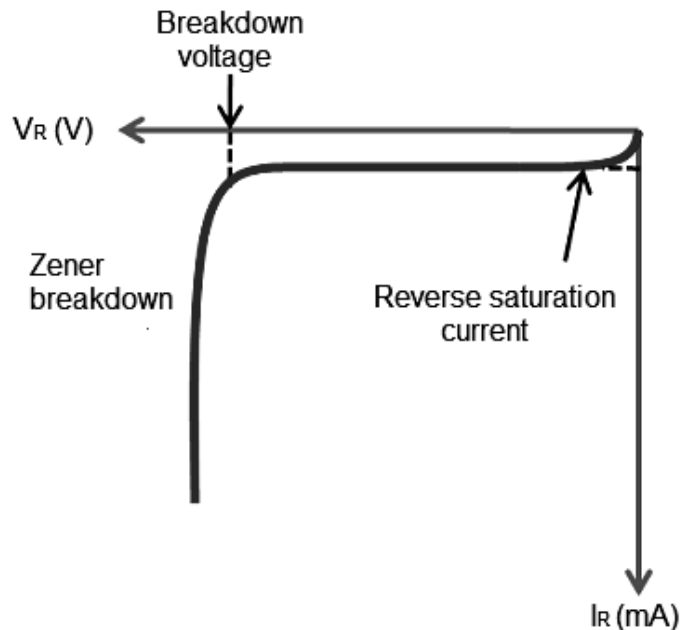


The valence electrons which gains sufficient energy from the strong electric field of depletion region will breaks bonding with the parent atom and become free electrons. This free electrons carry electric current from one place to another place. At zener breakdown region, a small increase in voltage will rapidly increases the electric current.

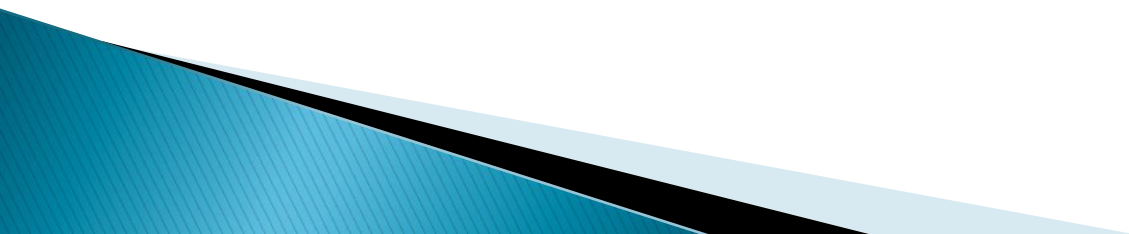


VI characteristics of zener diode

The VI characteristics of a zener diode is shown in the below figure. When forward biased voltage is applied to the zener diode, it works like a normal diode. However, when reverse biased voltage is applied to the zener diode, it works in different manner.



When reverse biased voltage is applied to a zener diode, it allows only a small amount of leakage current until the voltage is less than zener voltage. When reverse biased voltage applied to the zener diode reaches zener voltage, it starts allowing large amount of electric current. At this point, a small increase in reverse voltage will rapidly increases the electric current. Because of this sudden rise in electric current, breakdown occurs called zener breakdown. However, zener diode exhibits a controlled breakdown that does damage the device.



Advantages of zener diode

- Power dissipation capacity is very high
- High accuracy
- Small size
- Low cost

Applications of zener diode

- It is normally used as voltage reference
- Zener diodes are used in voltage stabilizers or shunt regulators.
- Zener diodes are used in switching operations
- Zener diodes are used in clipping and clamping circuits.
- Zener diodes are used in various protection circuits